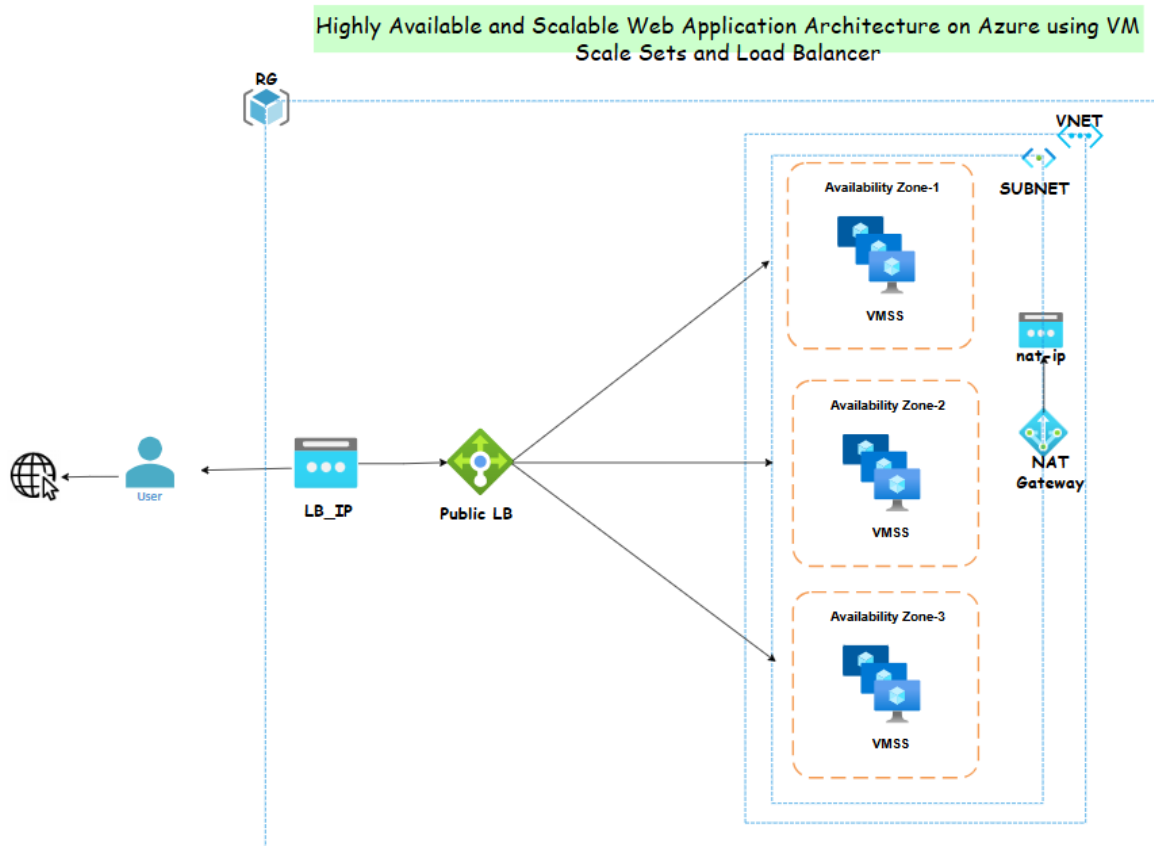


PROJECT 1: HIGHLY AVAILABLE AND SCALABLE WEB APPLICATION ARCHITECTURE ON AZURE USING VM SCALE SETS AND LOAD BALANCER

ARCHITECTURE DIAGRAM



STEP 1 — Create Resource Group

1. Azure Portal → **Resource groups** → Create
2. Name: rg-ha-web
3. Region: **Central India** (or any region with Availability Zones)
4. Create

STEP 2 — Create Virtual Network (VNet)

1. Portal → **Virtual networks** → Create
2. Resource group: rg-ha-web
3. Name: vnet-ha-web
4. Address space: 10.10.0.0/16
5. Subnet:
 - Name: subnet-web
 - Address: 10.10.1.0/24
6. Create

STEP 3: NAT Gateway Creation (Outbound Internet for VMSS)

- **What is NAT (Network Address Translation)?**

NAT allows private virtual machines (with no public IP) to access the internet securely using a shared public IP. In Azure, a NAT Gateway is used so VMs/VMSS can download updates and software without being directly exposed to the internet.

NAT is like a company receptionist phone number — all employees make outgoing calls using one common number, but outsiders cannot directly call individual employees

Purpose:

this enable outbound internet access for VM Scale Set instances so they can install packages (Apache, updates) when using a **Standard Load Balancer**.

Step 1: Create NAT Gateway

1. Go to **Azure Portal**
2. Navigate to
Network foundation → NAT gateways
3. Click **Create**

Step 2: Basics Tab

Fill in the following:

- **Subscription:** Select your subscription
- **Resource group:** rg-ha-web
- **Name:** natsub
- **Region:** Same as VNet (e.g., *Central India*)
- **SKU:** Standard
- **TCP idle timeout:** Default (4 minutes)

Click **Next: Outbound IP**

 Step 3: Outbound IP

1. Under **Public IP addresses (IPv4)**
→ Click **Create new**
2. Enter:
 - **Name:** nat-pip
 - **SKU:** Standard
 - **Availability zone:** Zone-redundant
3. Click **OK**

Click **Next: Networking**

Step 4: Networking (Most Important Step)

1. **Virtual network:** vnet-ha-web
2. **Subnet:** Select subnet-web

This attaches the NAT Gateway to the VMSS subnet.

Click **Review + Create → Create**

Why NAT Gateway is needed:

When using a **Standard Load Balancer**, VM Scale Set instances do **not get outbound internet access by default**.

NAT Gateway provides secure outbound connectivity, allowing VMs to install updates and packages (like Apache) without exposing them directly to the internet.

STEP 4 — Create Standard Public Load Balancer (ONLY basic shell, no backend yet)

1. Portal → **Load balancers** → Create
2. Resource group: rg-ha-web
3. Name: lb-ha-web
4. Type: **Public**
5. SKU: **Standard**
6. Click on **Add frontend IP configuration**

Frontend IP:

- **Name** : pip-ha-web
- **Public IP address** → Click create new
 1. **Name**: frontip
 2. **Availability zone**: zone Redundant
 3. save

7. Review and Create Load Balancer

Do NOT create backend pool yet

STEP 4 — Create VM Scale Set (IMPORTANT STEP)

This step automatically creates:

- VMs
- NICs
- NSG
- Links to Load Balancer

4.1 Create VMSS

1. Portal → **Virtual Machine Scale Sets** → Create
2. Resource group: rg-ha-web
3. Name: vmss-web
4. Region: same as RG
5. **Availability Zones** : Select **Zone 1, 2, 3** (if available) **THIS GIVES HIGH AVAILABILITY**
6. Scaling mode : **Manually update the capacity**: Maintain a fixed amount of instances
7. Instance count : 2
8. Orchestration: **Uniform**
9. Image: **Ubuntu 22.04 LTS**
10. Size: Standard_B1s
11. Authentication: Password

4.3 Networking (VERY IMPORTANT)

- Virtual network: vnet-ha-web
- Subnet: subnet-web
- Load balancing:
 - Select **Azure Load Balancer**
 - Choose **Existing**
 - Select lb-ha-web
 - Backend pool: **Create new**
 - Name: bp-vmss

4.4 Advanced Tab → Custom Data (Auto install web server)

```
#cloud-config
package_update: true
packages:
  - apache2

runcmd:
  - systemctl enable --now apache2
  - |
    cat <<'EOF' > /var/www/html/index.html
    <!DOCTYPE html>
    <html>
    <head>
      <title>Lucky Bakery</title>
      <style>
        body { background:#fff7e6; font-family: Arial, sans-serif; text-align:center; padding-top:100px; }
        h1 { color:#d2691e; font-size:50px; }
        p { font-size:22px; color:#555; }
      </style>
    </head>
    <body>
      <h1> Welcome to Lucky Bakery </h1>
      <p>Freshly baked happiness every day!</p>
    </body>
    </html>
    EOF
```

Review and Create → create

STEP 5 — Verify VMSS instances

VMSS → vmss-web → Instances

You **must see**:

- Instance 0, Instance 1
- Status: **Running**

STEP 6 — Configure Load Balancer (now the correct time)

Go to:

Load Balancer → lb-ha-web

6.1 Health Probe

1. Health probes → Add
2. Name: hp-http
3. Protocol: HTTP
4. Port: 80
5. Path: /
6. Interval : 5
7. Save

6.2 Load Balancing Rule

1. Load balancing rules → Add

2. Name: rule-http
3. Frontend IP: pip-ha-web
4. Port: 80
5. Backend port: 80
6. Backend pool: bp-vmss
7. Health probe: hp-http
8. Save

Open Port 80 in Network Security Group (NSG)

Purpose:

Allow HTTP traffic from the Load Balancer / Internet to VMSS instances.

Step 1: Open NSG

1. Go to **Network security groups**
2. Select NSG attached to VMSS NIC
Example: basicNsgvnet-ha-web-nic01

Step 2: Add Inbound Security Rule

1. Click **Inbound security rules**
2. Click **+ Add**

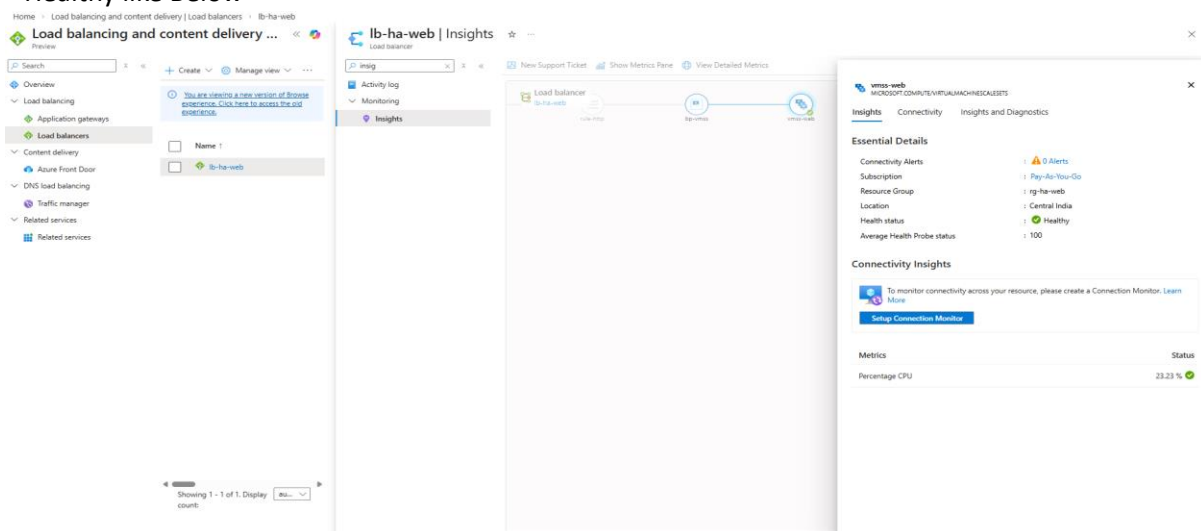
Configure the rule:

- **Source:** Any
- **Source port ranges:** *
- **Destination:** Any
- **Destination port ranges:** 80
- **Protocol:** TCP
- **Action:** Allow
- **Priority:** 100
- **Name:** allow-http

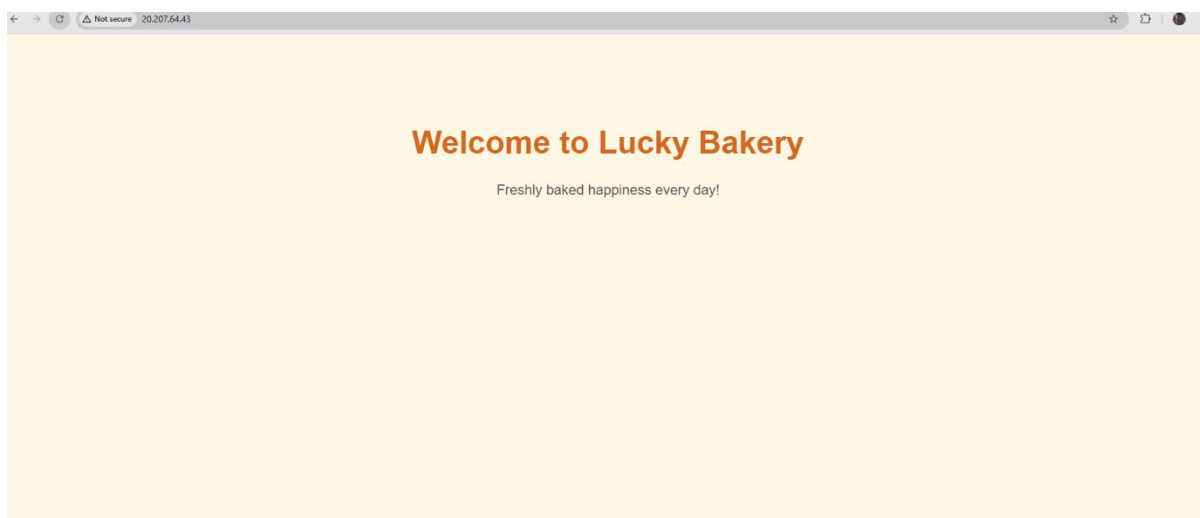
Click **Add**

TESTINGS

1. Navigate to Load Balancer → Type insights on the left pane. You should see the Health Probe Healthy like Below



2. Navigate to Public Ip addresses → click on frontip and browse it like <http://youripaddress/> should see the below page



3. Also you can see the detail metrics of the load balancer like below

